

DUMPSTER DIVER

Evaluation of Production Paths Forward



Brag & Bev LLC

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Approach to Finding a New CHEMIST



Universities / Colleges / Institutions

- University of Minnesota
- Georgia Tech (multiple departments, incl. Rodney Weber contact)
- North Carolina State University (multiple departments)
- Stony Brook University
- Virginia Commonwealth University (VCU)
- Purdue University
- University of California, Irvine (UCI)
- Central Michigan University
- California Polytechnic State University (Cal Poly)
- Texas A&M University (tamu.edu)
- Colorado State University
- Skidmore College
- University of California, San Diego (UCSD)
- Miami University
- University of South Florida
- University of New Mexico
- New Jersey Institute of Technology (NJIT)
- University of Michigan (multiple departments)

- Imperial College London
- University of South Carolina
- Washington University in St. Louis (wustl.edu)
- University of Texas
- Tulane University
- University of Maryland
- University of Maryland School of Pharmacy (separate department)
- Clarkson University (clarkson.edu)
- University of Denver (du.edu)
- University of Edinburgh (ed.ac.uk)
- University of Washington (uw.edu)
- University of Utah (utah.edu)
- University of California, Los Angeles (UCLA)
- University of Warsaw, Poland (uw.edu.pl)
- Virginia Tech (vt.edu)
- University of North Dakota (und.edu)
- Lund University, Sweden (lth.se)
- University of Iowa
- Penn State University

Companies / Labs / Other Institutions

- **Freelance Formulations**
- **Beakers and Bottles**
- **ARE Labs**
- **ATS (Aerosol Technical Solutions – via John Chadwick)**
- **Uno FormulationsDesign.lth.se (likely tied to Lund University design group)**

Path 1: DIY Formulas => MB Industries (BOV)

- **Process:** Source our own chemicals, mix/test small 1L batches, ship to MB Industries, Pennsylvania => MB fills into BOV cans => they send back units to test.
- **Regulatory:** No VOC concern; BOV is air-powered.
- **Cost:** Approximately \$1,000 for chemicals and supplies for initial formulation batches.
- BOV is usually 20–50% more expensive than a comparable traditional aerosol.
- On a per-unit basis, depending on volume:
 - Traditional aerosol filling: \$0.15–0.30 extra cost per can.
 - BOV filling: \$0.40–0.60 extra cost per can
- **Pros:** Full control of formulation, BOV = eco-friendly marketing edge.
- **Cons:** Slower, more bench work on my end, even with the DIY could be more costly in the long run.



Aerosol Research and Engineering Labs



4 DIY Formulations

F1 — Baseline Foaming Cleanser — 1 L Batch Card

Goal: Simple, clear, stable; good foam; VOC-free; BOV-ready. | Target pH: 5.8–6.0 | Viscosity: »250–400 cP

F2 — Boosted Cleaning (Hydrotrope + Co-solvent) — 1 L Batch Card

Goal: Better grease/soil tolerance; remains BOV-sprayable. | Target pH: 5.8–6.0 | Viscosity: »250–450 cP

F3 — Heavy-Duty Anionic System — 1 L Batch Card

Goal: Stronger cleaning; clearer rinse; anionicnforward. | Target pH: 5.8–6.0 | Viscosity: »300–600 cP

F4 — Advanced (Builders + Polymer Anti-redeposition) — 1 L Batch Card

Goal: Highest performance; watch clarity/viscosity & filtration. | Target pH: 5.8–6.0 | Viscosity: »300–600 cP



Path 2: SprayProducts (Conventional Aerosol)

- **Process:** SprayProducts handles formulation + nozzle pairing + filling.
- **Regulatory:** VOC compliance not a problem (their expertise).
- **Notes:** They emphasized confidence, attitude, and alignment with our vision.
- **Pros:** Fastest to manufacturable solution, strong technical partner.
- **Cons:** Standard aerosol format (less eco-friendly than BOV), VOC compliance still applies.



Overview

Path	Partner	System	VOC Compliance	Cost Impact	Speed	Notes
DIY + MB Ind	MB Industries	BOV	No Concern / Filled with Air	+ \$0.15-\$0.30/can	Slowest	You mix, they fill
SprayProducts	SprayProducts	Aerosol	Not an issue	Standard	Fastest	Very confident in their capabilities

Fork in the Road

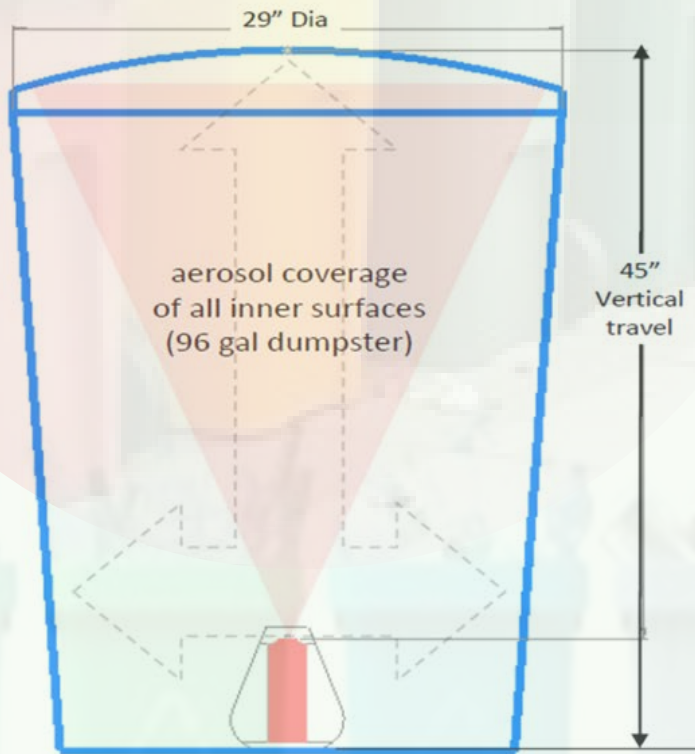


Formula-1 Cleaning Agent

- Cleaning agent that creates foam upon activation
- Foam is designed to adhere to the sides and surfaces of the trash bin, loosening grime, grease, and debris, neutralizing odors.

Formula-2 eco-friendly SAP materials Rinsing Agent (Post-Cleaning Residue)

- The rinsing agent is released after the cleaning agent is activated and adheres to the trash bin surfaces.
- Designed to absorb moisture and bind debris left behind by the cleaning agent.
- Solidifies into a residue that is easy to remove



Aerosol contents-

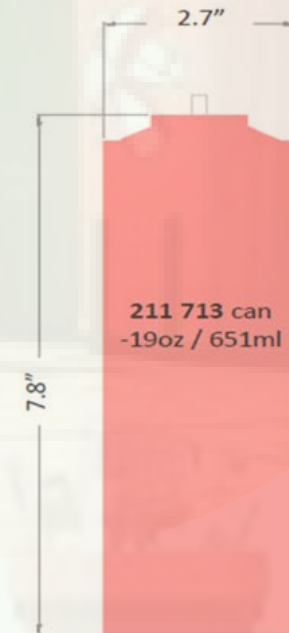
Cleaning/Foaming agent (hard surface cleaner)

- not compatible as a vertical upward spray.

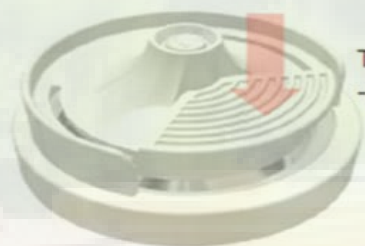
Fogger? can be sprayed to linger in air

- more compatible as upward spray.
- Does not 'clean' anything.

Propellant- required for spray distribution force



Aerosol formula and canister
Requirements and constraints
Draft-4 8/11/25



Total Release nozzle
- downward force to trigger nozzle actuator

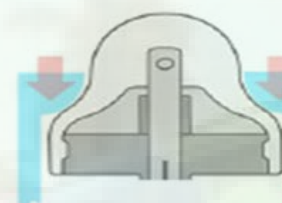
Option1
Break away pull tab
for safety lock

Housing
- Impact trigger is part of housing

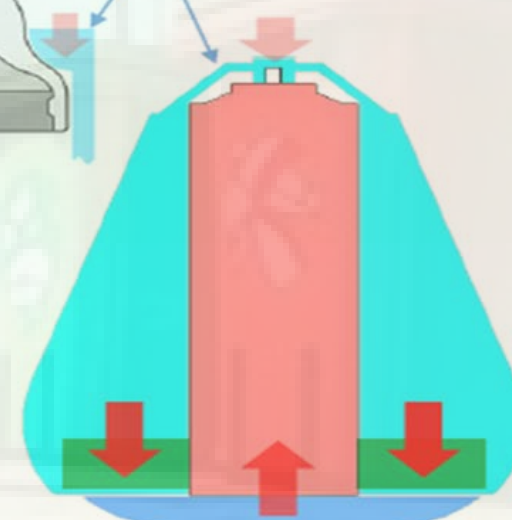
Weight
- Self-righting
- Gravity assist for
impact triggering

Option2 Break away pull tab

Bottom base
- Connected to spray canister



Weight of housing
- downward force to trigger nozzle actuator



Impact resist
after hitting bottom of bin

Trigger mechanism concept
Draft-1 8/4/25